# Damien House - CTO & VP Engineering - Capability Statement

## Introduction

Organizations can prototype faster than ever in the agentic era, but speed alone is not a strategy. Without governed architecture, clear context, secure delivery systems, and durable operating practices, rapid experimentation rarely becomes durable execution. To address that challenge, Damien House evolved 109.studio's practice toward applied AI, governed workflows, and production-grade systems.

As a fractional CTO, senior engineering advisor, and hands-on principal architect, Damien helps organizations move from concept to production with clear architecture, governed workflows, and durable systems built to scale. Drawing on 20+ years of platform modernization, DevOps transformation, Drupal architecture, and executive technical leadership, he combines seasoned delivery discipline with active R&D in Agentic Engineering, Agent Experience (AX), retrieval-backed context systems, AI-enabled content operations, and human-in-the-loop automation.

For non-technical founders, Damien acts as translator, safety net, and execution partner. He helps teams avoid bad technical bets, move from prototype to production with less risk, and build systems that can scale beyond MVP. For technical leaders, he brings architectural rigor, governed AI workflows, stronger DevEx and DevSecOps practices, and a practical framework for introducing automation without chaos. The result is governed intelligence built on proven engineering foundations.

## Services Overview

- **Fractional CTO & Technical Strategy:** Damien provides executive technical leadership for founders and growth-stage teams that need more than tactical development. He translates product ambition into architecture, sequencing, delivery governance, and risk-reduction decisions — helping organizations move from vague ideas and unmanaged experimentation to buildable, scalable systems.
- **Prototype-to-Production Architecture:** Damien turns early concepts, prototypes, and AI-generated mockups into versioned specifications, roadmap decisions, and production-ready system designs. Using Agentic Engineering and artifact-first execution, he compresses planning and implementation cycles while preserving architectural integrity, accessibility, and review discipline.
- **Agentic Product Architecture & AI Governance:** Damien designs the substrate AI systems need to operate safely in production. That includes explicit action surfaces, context layers, scoped permissions, retrieval-backed knowledge, review paths, and human-in-the-loop controls. Rather than developing simple wrappers, he helps organizations build governed systems that can support meaningful automation without exposing the business to avoidable risk.
- **AI-Enabled Content Operations & Governance:** Damien helps organizations apply AI to content estates where manual audits, accessibility remediation, metadata generation, and editorial QA do not scale. He designs governed content workflows that improve audit coverage, reduce manual review burden, strengthen accessibility performance, and make large content environments more maintainable.
- **Engineering Leadership, DevEx & Platform Modernization:** For organizations with existing teams and legacy platforms, Damien serves as a principal-level architecture and delivery partner. He modernizes infrastructure, standardizes environments, improves CI/CD and documentation, strengthens security and code-quality workflows, and leaves teams with a stronger technical foundation for safer releases and future automation.

## Methods and Tools

- **Design Science + Agentic Engineering:** Engagements begin by clarifying the problem, the user consequence, and the success condition before implementation starts. Damien combines design-science thinking with agentic engineering's artifact-first execution model so every sprint produces not only working software, but also durable context: specifications, decisions, interfaces, and operating assumptions that keep implementation coherent as complexity grows.
- **Agent Experience (AX):** Damien treats AI agents as operational participants that need explicit guidance, not magical autonomy. He designs action, governance, and context surfaces so agents can assist productively with development, operations, and content workflows without improvising against hidden state or production systems.
- **Retrieval, Context Control & Organizational Memory:** Damien uses retrieval-backed context systems, structured knowledge, and governed memory patterns to improve consistency, reduce hallucinations, and make AI more useful across engineering, product, and content operations.
- **Cloud-Native Delivery, DevEx & DevSecOps:** Damien improves delivery systems through standardized environments, CI/CD, automated testing, AI-assisted code review, security scanning, and documentation practices that reduce operational friction while improving reliability, maintainability, and release confidence.
- **Human-in-the-Loop Content and Workflow Automation:** Damien applies AI where it creates practical leverage — including content auditing, accessibility support, metadata generation, and workflow automation — while preserving human review where quality, judgment, or compliance matter most.

## Sample Project Types

The following are representative categories of projects and initiatives Damien leads, reflecting his expertise and the R&D areas he actively explores:

- **Full-Stack Platforms, Modernization & DevOps Automation:** Damien designs scalable web platforms and modernizes legacy systems with an emphasis on reliability, maintainability, and delivery speed. His modernization work combines architecture planning, cloud migration, environment standardization, and AI-augmented execution patterns that can compress traditional 3-to-6 month migration efforts into 3-to-4 week cycles; in comparable enterprise delivery, this style of approach has reduced a 3-month multi-developer effort to 1.5 months. The result is faster modernization without sacrificing operational control.

- **AI Agents, Retrieval Systems & Intelligent Automation:** Damien develops AI-augmented features and agentic workflows that connect models to governed context rather than loose prompts. He uses retrieval-backed search, tool-calling patterns, and structured workflow automation to improve answer quality, reduce dead ends, and make AI useful inside real operational systems. This includes semantic search that improves answer accuracy by 40% to 60% and cuts no-result experiences by 50%, plus agentic workflows that support content creation, knowledge retrieval, service orchestration, and human-in-the-loop decision support.

- **Developer Tooling, Content Intelligence & Open Source Ecosystems:** Damien builds reusable tools and delivery patterns that improve developer productivity and editorial scale. One example is AI-driven diagnostic and audit work that expands content assessment from manual 50-page samples to full-estate review across 6,000+ content items, reducing a massive enterprise audit to roughly $400 in API compute instead of months of manual labor. He applies the same philosophy to quality engineering and editorial governance, using AI to reduce repetitive toil while keeping humans in control of final decisions.

- **Cross-Disciplinary Product R&D:** Damien leads exploratory projects that validate ideas before organizations overinvest in the wrong build. By pairing design science, architecture strategy, and AI-assisted prototyping, he helps teams test concepts quickly, gather evidence, and move from novelty to operational value. This R&D lens allows him to distinguish between AI features that merely generate output and those that measurably reduce delivery time, lower audit cost, improve accessibility, or increase system reliability.

## About Damien House

Damien House is the Principal and Founder of 109.studio, based in Washington, DC. He has built a distinguished career as a software architect and engineering leader, delivering successful outcomes across the private sector, government, and startup world. Over the course of more than two decades, he has led engineering teams and projects of all sizes – from revamping federal government web platforms used by millions, to advising lean startups on cutting-edge AI product development. His work with the U.S. Department of Agriculture and International Trade Administration, for example, modernized mission-critical systems and introduced first-of-their-kind DevOps and cloud practices in those organizations. In the startup arena, he has guided founders through technical maze of launching new products, like the AI operations platform where he served as a Fractional CTO.

What truly defines Damien is his systems thinking and continuous growth mindset. He holds a formal background in both computer science and design and continually updates his expertise – whether by experimenting with the latest AI development tools or pursuing professional certifications (such as in UX design and cloud architecture). This cross-disciplinary foundation enables him to operate fluidly between the worlds of software engineering, product design, and business strategy. He approaches problems holistically, considering human factors, process dynamics, and technical architecture as parts of one system. Colleagues and clients often note his enthusiasm for *learning in public* – Damien frequently writes, speaks, and shares lessons learned, believing that knowledge should be open and iterative. This habit not only keeps him at the leading edge of industry trends but also demonstrates the transparency and thought leadership he brings to every engagement.

As a leader, Damien is approachable, pragmatic, and vision driven. He can dive deep into code reviews or infrastructure debugging with engineers one moment, then articulate a clear technology roadmap to executives the next. He thrives on collaboration and has a talent for uniting cross-functional teams around a shared technical vision. By fostering an environment where creativity is coupled with accountability, he consistently delivers solutions that are inventive yet reliable. Clients of 109.studio find in Damien a partner who is as invested in their mission as they are – a CTO-level ally who will roll up his sleeves to solve problems today *and* architect for tomorrow's opportunities.

In summary, Damien House stands out as a technology leader who bridges the gap between innovative idea and impactful implementation. His agency's services, methods, and ethos reflect a rare blend of engineering maturity and creative innovation – offering startup founders and established organizations alike the capability to build the future with confidence and clarity.